PART 2

PROBLEM SOLVING STRATEGIES: AN ENGINEERING CULTURE

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6

THE EVOLUTION OF DESIGN PROBLEMS

For when the One Great Scorer comes to mark against your name,
He writes – not that you won or lost – but how you played the Game.
Grantland Rice, American sports writer in 'Alumnus Football'(1941)

We now focus on some general issues about the whole process of designing. In doing so, our presentation needs to be placed in a context of the designer's task mix. Almost all design is done by humans, who take on various roles during the management of the whole design task. This part of our discussion makes specific reference to those aspects of the design activity that are (currently) still the exclusive domain of the human mind.

We are being somewhat pedantic in implying that not all design is performed by human beings. In the context of *artificial intelligence* (*AI*) there is a substantial school of thought that deals with *automated design*. It has long been the dream of AI practitioners that once the requirements and design boundaries have been determined, a computer programme can intelligently resolve most, if not all, conflicts in a design. In the introductory chapter to *Intelligent Design and Manufacturing* (Kusiak, 1992), Kannapan and Marshek write:

"...substantial research effort in the area of engineering design has been directed at significantly improving the productivity of the design process."

This statement clearly identifies the process of design as requiring some improvement in support tools so that the human designer may be more productive in the delivery of design results. Some of these tools are referred to as computer aided design (*CAD*) or decision support systems (*DSS*). *CAD* is often interpreted as the process of preparing virtual design embodiments by solid geometric modelling programmes (for example *AutoCAD*, *Pro/ENGINEER*, *UnigraphicsCAD*, *CATIA* and *SolidWorks*, to name just a few of the geometric modelling products available). However, computer support has a far wider influence on design than that encompassed by mere geometric modelling. Kinematic and dynamic modelling of engineering systems, thermo-fluid modelling of complex thermal and flow processes, and even process modelling of robot based manufacturing offer substantial support for the designer beyond the embodiment activity. Sophisticated search and logic programmes (sometimes referred to as *search engines* and *inference engines*) provide help with resolving complex interactive design requirements. Yet in spite of the available, impressive, array of tools for computer based design, they can not be expected to replace the most important *design-front-end* activity of the human designer. What is invariably the commencement point of any design is the *recognition* that a *need for design* exists. Once the *need* has been identified, many *conflicts* must be *negotiated* through to some comprehensive statement of *objectives* and *requirements* of the ultimate design (the result of the design process), and the way we might measure the succes of the result (*design criteria*). It is this critically important human-directed aspect of designing that ultimately determines the majority of the features of the design result.

Chapter 6 briefly introduces the various elements of the design process, and Chapter 7 presents some useful decision making tools. We introduce the notion that design is *not a science*, but a *discipline*, where there are *no easily refutable conjectures*. As a direct result of this notion, we can not suggest that the design approach we propose is mandatory for best results. On the other hand, in what follows we try to develop a disciplined way of thinking about the general process of design, specifically useful for inexperienced designers. While our approach in Part 1 followed a slightly *deconstructionist* approach to design for structural integrity(*structural distillation*), Part 2 follows a

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strongly *constructionist approach* to design. We still need to *see problems through the designer's eye*. However, here we need to see the *whole* as it becomes *constructed from the several parts* rather than the other way around.

6.1 Cultural development of the designer

"Without confidence in its own culture, a nation does not seriously exist." Donald Horne[6.1]

Author and academic, Professor Donald Horne, was referring to a country's cultural awakening, but his comment may equally well be understood in terms of a growing interest in the design and development of high quality engineered products. Almost half a century ago, media guru, Marshall McLuhan, wrote (McLuhan and Fiore, 1967):

"Ours is a brand-new world of allatonceness. 'Time' has ceased, 'space' has vanished. We now live in a global village...a simultaneous happening."

McLuhan was referring to the psychological effects of globalised media. Even he could not have foreseen how globalised media would impact on manufacturing competitiveness. Manufacturers can directly access suppliers' databases at the *"flick of a keyboard"* anywhere in the world, day or night. Consumers are constantly exposed to internationally demanded standards of product quality. Many large manufacturers can now source manufacturing technology, and have direct access to product quality management tools, from anywhere in the world. These factors are largely responsible for such products as the MDX helicopter (see Figure 6.2), fuselage made in Australia, engine in England, gearbox in Israel. Also, many manufacturers of consumer products supply parts and even whole assemblies to a wide range of market outlets. The term for some of these products is *"badge engineering"*. Fashion houses like Calvin Klein and Gucci don't manufacture watch movements, yet they market watches.

In such an agressive atmosphere of marketing no manufacturer can now afford to spend five years in cradle-to-launch activities (typical time period for new models of automobiles). Where products involve computer technology, there is an even greater urgency to deliver to market. Most computing chips have a useful technology advantage for relatively brief periods of the order of one or two years.

With many consumer products, the manufacturer's competitive advantage (*edge*) results from *agility* (the capacity to rapidly respond to market demands). For products that rely on significant specialist engineering expertise, such as hydraulic turbines, turbo-alternators, or aerospace products, the competitive *edge* results from *flexibility* (the capacity to respond to a wide range of alternative design requirements).

Designers have significant influence in both aspects of manufacturing, *agility* and *flexibility*. In addition, quite apart from its marketing significance, a well designed and manufactured product can act as a source of national pride and a driving force for developing a strong and vibrant *design culture*. In this culture we find that *the disciplines of engineering design* and *industrial design* appear to have developed separately but in parallel.

The Encyclopaedia Britannica describes industrial design as:

"...design of products made by large scale industry for mass distribution. Designing such products means, first, planning their structure, operation, and appearance, then planning these to fit efficient production, distribution, and selling processes. Clearly, appearance is but one factor in such a complex process. Nevertheless, in consumer goods especially, appearance design is widely accepted as the principal virtue of industrial design; it is that portion of the whole least subject to rational analysis and, like craft secrets of the past, most advantageous in commercial competition. On the other hand, design of equipment for production, for services, and for sports is expected to demonstrate utility, but in these products, too, appearance design is increasingly important."

This is a highly simplified description of what industrial designers do, and, in essence, it is not all that different from what we might expect that engineering designers do. Perhaps the best way to describe the difference and the commonality between the two disciplines, industrial and engineering design, is to focus on the considerable specialist and organisational skills each discipline needs to encompass. The industrial designer's specialist skills might be seen to lie in the visual, ergonomic and material selection aspect of design. The engineering designer specialises in technical analysis, mathematical modelling and structural integrity issues. In many design situations there is significant overlap in the task mix of both types of designers.

6.1 Australia's Best:A Digest of Australian Achievement, 1988, The Australian Design Council.

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6.1.1 Engineering design as wisdom

Engineering designers have been struggling with the definition of the discipline since 1966, when the first conference on engineering design was held in England (Gregory, 1966). The discipline is supported by a substantial body of literature (see the *Annotated bibliography of engineering design* in our reference list), yet most institutions teaching engineering design do not have a prescribed text for supporting the presentation of general design procedures. Many institutions feel compelled to offer their own versions of design to undergraduates, including a combination of guided design exercises and lecture material. It is commonly hoped that the guidance in the design exercises will convey and instil some elements of the general design process.[6.2] Our experience is that application of the general design principles is heavily infuenced by the technology context within which the project is set (Samuel and Weir, 1991). This process of building contextual or domain-specific experience has been referred to by French as *received wisdom* (French, 1988).

Gaining *wisdom* in engineering design, whether domain-specific or general, is a process that is low in procedural content and is often difficult to articulate. We quote McLuhan again[6.3], whose highly appropriate description of media is equally applicable to the formal design process as we proceed from *need* to *design objectives*:

"Environments are invisible. Their ground rules, pervasive structure, and overall patterns elude easy perception."

This is the portion of the whole design experience *least subject to rational analysis.* Broadly speaking, the engineering design discipline is an embodiment of current engineering culture. In that sense, it encompasses the current practices, ethics and linguistic character of all that exists and all that has gone before it in engineering. The essential objective of this book is to reveal and highlight the key features of that culture to young engineers.

6.1.2 Engineering design as discipline

First of all we note that engineering design is a discipline, with disciples who follow and ultimately disseminate the discipline to others. Consequently, the discipline is a lifelong learning experience and engineering design is as much about the practice of engineering as it is about the education of engineers to deliver that practice in the most effective way.

The engineering design discipline is often sharply contrasted with the engineering sciences, which are doctrines, rather than disciplines. These doctrines commonly presume some model of reality to enable them to deliver their predictions of behaviour along the lines of their current orthodoxies. Engineering science develops along the rigorous paths of the well recognised scientific method, involving model building and experimental verification. Engineering science looks for easily refutable (testable) rules for predicting model behaviour. Unless a rule can be tested, and refuted if false, it has no place in engineering science.

Engineering design is a problem finding and problem solving procedure. While it is strongly based in information management, it is not science. Nevertheless, we make consistent and substantive use of engineering science models for problem solving. Moreover, our problem solving methods follow closely the procedures used for scientific problem solving. These procedures have at various times been identified as *the design method.* Using a *design method*, or a prescriptive procedure for managing design information does not constitute *design science*. There is a sharp difference between our design discipline and the doctrinaire sciences. While we often use the design method for problem solving, it represents a *sufficient*, though *not necessary*, procedure. There is no way of verifying (or refuting) the proposition that *only the design method leads to useful design solutions.* What, in fact, can we identify as *useful design solutions?* What are the dimensions of a *good design?* These and associated issues are the subject of this second part of our book.

Both engineering science and engineering design are living organisms that respond to changes in technology and philosophy. However, the lead times for changes in the design discipline are, of necessity, shorter than the lead times for doctrine changes in science. As an example, the change from Newtonian mechanics to relativistic mechanics took several hundred years, while the change from horse-drawn vehicles to the automobile took less than 50 years. In printing the change from manual typesetting has changed to electronic typesetting in about 20 years.

6.2 A strong reminder of the older style of master-apprentice form of engineering training

6.3 McLuhan and Fiore (1967), loc. cit.

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The culture of engineering science is rich in procedure. Thermo-fluid science relies heavily on the Navier-Stokes and Bernoulli's equations, and the various *laws* of thermodynamics. Mechanics of solids makes consistent use of equations based on energy conservation. While we have no such well established rules in engineering design, there are some *design-support* procedures that result in credible and effective design problem management. Clearly, the design discipline has its origins in an artisan culture. Great engineering designers of the last century such as Stephenson, Watt, and Brunel had no formal training in design, but rather acquired their considerable design skills and engineering wisdom amidst the apprenticed artisan culture of the day. Nowadays, maintenance of an informal artisan character in design would serve to relegate the discipline to one seriously lacking in scholarship. However, the benefits of acquiring experience and judgement through extended association with a "master practitioner" should not be dismissed.

It is instructive to examine the experience of Michael Faraday (1791-1867), widely regarded as the *father of electricity*. Of necessity, Faraday left school at age 13. At age 22 he was taken on as Sir Humphrey Davy's bottle washer and commenced his own experiments at age 29. Davy (1778-1829), a chemist with a considerably more affluent background than Faraday, was by this time the director of the Royal Institution. Consequently, Faraday was exposed to the substantial wisdom of Davy for seven years before commencing his own experiments. Faraday's genius could be easily mistaken for the triumph of *nature over nurture*, thereby ignoring the beneficial impact of formal training.

The Wright brothers had no more formal schooling than did Faraday. Yet they were able to triumph in the design and construction of flying machines, when many formally trained engineers and scientists failed before them (Kelly, 1996). In fact, they conducted carefully planned experiments and made extensive use of the *received wisdom* of their trade art in bicycle manufacture.

Perhaps the most significant issue in the development of a credible design culture is the recognition of the need for formal training in design *discipline*. Here we refer not to the practice of design, but to the carefully controlled application of its procedures.

6.2 Design problem solving

In his book, *Invention and evolution*, Michael French (1988) describes the design requirements of a humble clothes peg. In 1988 he claimed this to be an unsolved problem, but his description of the design detail in this apparently simple device we all take for granted, is a classic example of the richness of design studies. French in his description of the design problem did not suggest that there were no *useful embodiments* for clothes pegs, but that all these embodiments are the result of manufacturing compromises that leave *many of the original design needs unfulfilled*.

Another example of difficult consumer product design is the bayonet fitting on household electric lights. The connection between the light globe and the socket is made via two, low melting point, metal contacts on the rear of the globe. Unfortunately the heat from the incandescent globe softens these contacts, and the spring loaded socket contacts embed themselves in the soft metal. When the globe burns out, it becomes almost impossible to remove. To make matters worse, on the light fitting the *handedness* of the screw securing the socket to the fitting is invariably the same as that of the bayonet fitting. Hence, when trying unsuccessfully to unscrew burnt out globes, one inevitably unscrews the whole socket assembly (refer to Figure 6.1).

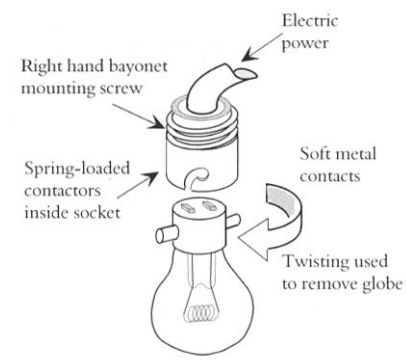


Figure 6.1 Household electric light globe fitting

Design involves both the intellectual and emotional substance of the designer. Intellectual because designing requires considerable analytical skill. Emotional because the designer is a human, designing for other humans, and this process brings with it a vast *grab-bag* of issues the designer

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must take into account. We refer to the environmental, social, aesthetic and economic nature of design. The designer must be *aware* of these matters to allow modification of the designed product accordingly. A key feature of a good designer is the ability to consider many points of view and to arbitrate between them to the overall benefit of the designed product. Another key feature of good designers is their capacity for coping with change (*flexibility*). In executing design, the designer must progress through an ever changing tableau of ideas and conflicts that bear on the ultimate outcome. This is true even in such simple, familiar products as the clothes peg and the light fitting.

Petroski (1996) presents the design process through the experiences of the inventor, identifying no less than 10 distinctly different patents for the simplest of engineering artefacts, the paper clip. As French (1988) has noted, *"design is a rich study"*.

As a first step in the development of design skills we need to characterise the substance and extent of design. There are at least three components to this characterisation. The first of these is *the designer*. We recognise that the designer is mainly human, although, as noted earlier, there is considerable and rapidly growing interest in artificial intelligence as a means of automating design (Brown et al., 1992). Hence we can argue that, at least in some situations, the designer may be *significantly aided by a machine*. Furthermore, the designer is often a *group*. When group responsibility is taken for a given design, the dynamics of team interaction come into play. Under these circumstances the team leader must be not only aware of the nature of these dynamic interactions, but capable of handling *conflict negotiations*.

Evaluation of psycho-social interactions involved in team dynamics is outside the scope of this text. Nevertheless, we need to recognise that engineering product design is rarely the domain of the individual. Perhaps the most challenging feature of participating in a design team is the individual ambition to add value to the ultimate result. The task of the effective team leader is to extract the best possible value from each individual in the team.

The second, equally important, component of design is the *procedure adopted for design*. Although this procedure is not rigidly defined, many of its steps are repeated in every design. In a formal way this is referred to as *the design method*. We also refer, among design practitoners, to the design method as an essentially *transportable property* of all design experiences. When reviewing case examples in design, it may be very tempting simply to adapt the procedures used for designing solutions to a problem *similar to our problem*. This is a false interpretation of the objectives of case examples, whose intent is to focus not on the *destination* but on the *journey*.

The third component of design is *the product* we are designing. Designers are called upon to address a vast range of products, but at this early stage of our discussion we do not focus on any specific product. Suffice to say that not all products are designable by all designers. Each product must be seen in its own context of application, since there is a considerable range of ideas and conflicts which are peculiar to the operating environment of a specific product (*domain-specific*). Moreover, experience and growing expertise in the design of specific products brings with it a kind of wisdom not easily transported to other domains of product design. Typically, specialist expertise in the design of rock-crushing machinery or steam engines has little relevance or transportability to jet engine or electric alternator design.

6.2.1 The designer as a person

Any discussion of the human designer must necessarily identify the skills possessed by the designer and how these skills are to be developed. We describe the behaviour of the human engineering designer in the context of design. We seek to distinguish those characteristics of the designer that influence the process of designing. This will enable us to recognise characteristics we need to develop in becoming good designers.

Engineering design is a relatively young discipline. James Watt (1736-1819), the inventor of the condenser for steam engines, was a land surveyor, colliery superintendent and businessman. Eli Whitney(1765-1825), the inventor of the cotton gin, was responsible for the modern concepts of mass production. Isambard Kingdom Brunel (1806-1859) designed and built railways, bridges and ships, in addition to working as an architect. Watt, Whitney and Brunel, among other great engineers, clearly performed tasks that are currently regarded the domain of engineering design. Yet these activities in the late 18th and in the 19th centuries were simply included into the broad general role of engineering. With the

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evolution of a great diversity in manufacturing technology, the artisan-engineer evolved into the manufacturing specialist and design specialist.

Coupled with the evolution of diversity in manufactured goods there has been a consistent evolution and development of specialist experimental and analytical tools in engineering, demanding some highly specialised technical skills. The engineering designer can not be expected to be equally conversant with all of these various specialties. Nevertheless, we must be sufficiently informed about them to identify those aspects of the design where they apply and to link the proper technology to these aspects of the design.

As we will demonstrate, the most significant skill of the engineering designer is the capacity for processing large quantities of diverse information in a well ordered way. This skill becomes especially useful when evolving technical problems from loosely identified needs. Often, many of the requirements of a design are unarticulated by the designer's clients. In general, the designer must identify the various functions which fulfil the requisite need with minimum compromise. Judgement in identifying the best (or at least a satisfactory) set of compromises, is part of this collection of design skills.

We add to these attributes a type of wisdom that regards all design problems, from the apparently trivial to the complex, as proper challenges for the designer's skills. This wisdom is related to the associative skills of *receiving*, *responding* and *valuing*. Designers face many challenges in their lives, and responding successfully to problem solving is a confidence-building experience. By receiving, responding to, and valuing each design challenge without *fear or favour* , the designer will invariably develop this confidence.

It is hardly accidental that this introduction to the *designer as a person* has the flavour of a motivational message. Much of the emotional makeup of the designer will develop in professional life, sometimes years after graduation. Our intention in dealing with it at this early stage is to set the scene for all the learning experiences that follow.

6.2.2 What constitutes design?

Before a discussion on how to go about designing products or systems, we need to explore what designers do. In other professions, such as the law or medicine, we can easily describe the activities of the participants. Doctors heal and lawyers litigate. Designers on the other hand find it singularly difficult to articulate the boundaries of their domain of influence. A few examples might help to illustrate, as well as deal with, this difficulty.

(a) Designing a better helicopter

Figure 6.2 is an artist's impression of the McDonnell-Douglas MDX twin turbine helicopter. In order to achieve a substantial *marketing advantage*, the design goal for this passenger transport aircraft called for nearly twice the payload-to-power ratio available from conventional helicopters. A wise combination of several conceptual design factors contributed to the achievement of this ambitious design goal. The main rotor blades have a flexible polymer mounting at the rotor hub. This arrangement, resulting from a French patent, significantly reduces the mechanical components used in the blade angle adjustment mechanism.



Figure 6.2 McDonnell-Douglas Explorer helicopter

The cabin and empennage (stabilising arrangement of the tail section) were to be constructed in carbon-fibre reinforced polymers. In aircraft construction, combining polymers and conventional aluminium panels is fraught with considerable risk in manufacturing and airworthiness testing. However, the weight saving was seen as a challenging goal, but achievable given the company's experience in the design of military

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aircraft that had already been built using this material combination.

The aircraft was designed without a tail-rotor. This feature is referred to as the *notar* design. The torque on the main rotor blades imposes an equal and opposite torque on the helicopter cabin. To overcome this problem, most conventional helicopters are fitted with a tail-rotor whose plane of rotation is approximately perpendicular to the rotation plane of the main rotor. Appropriate adjustment of the blade-angles on the tail rotor balances out the torque component imposed on the cabin by the main rotor.

In the *notar* design, the conventional tail rotor is replaced by a fluid dynamic behaviour called the *Coanda effect*, named after its discoverer, Henry Coanda (1886-1972), a Romanian aviator. This effect exploits the capacity of an air jet to *cling* to a curved wall. Figure 6.3 shows the operation of the *Coanda effect* schematically. Air jets are blown out on the side of the tail boom, resulting in a local low pressure region and net thrust on the boom.

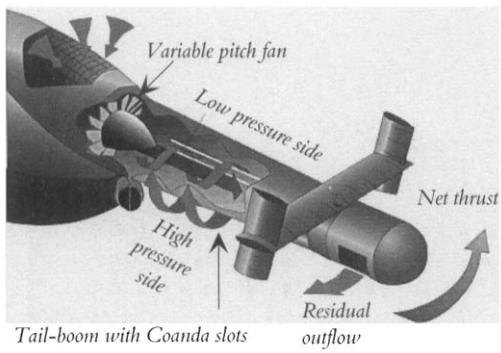


Figure 6.3 Coanda effect on Notar helicopter tail boom. (Courtesy of Kulikov Aircraft Co.)

The *notar* design allows the aircraft to approach buildings and people more closely than is possible with conventional helicopters. Additionally, the weight saving resulting from the absence of the tail-rotor drive and blading is substantial.

The MDX is clearly the result of substantial *know-how* (*domain-specific wisdom*) about helicopter design. However, marketing staff at McDonnell - Douglas had to establish a *need*, or a market opportunity for this design. Once established, the need was serviced by a creative cooperation of experience and design skill. Experience with composite manufacture, polymer mounted main rotors, and the *notar* design, were essential elements of this design synergy. The final result was also critically influenced by the broad range of design skills available to McDonnell-Douglas.

In summary then, this design problem involved:

- identification of a *need* to be serviced, namely the need for a more effective civilian helicopter service: this was the *design goal*;
- evolution of *technical design objectives* and the technical design limits, or *design requirements* within which the design embodiment should be achieved (examples here are : maximum cruise speed at sea level ≥ 278 km/hr; maximum endurance ≥ 4 hr);
- creation of *solution alternatives* from which a final design embodiment would be chosen;
- focused *decision making* on the final design choice;
- delivering clear and unambiguous *design specifications* to the manufacturer: this would include, among other things, drawings, airworthiness testing procedures and material testing specifications.

(b) Designing a better mousetrap

Ralph Waldo Emerson (1803-1832) American philosopher and poet, is credited with the following quote:

"If a man write a better book, preach a better sermon, or make a better mouse-trap than his neighbour, tho' he build his house in the woods, the world will make a beaten path to his door."

The *standard* mouse-trap has been in use for a considerable time. One current form of its embodiment is shown in Figure 6.4. At first sight, it may be surprising that anyone would want to improve on such a simple and apparently effective device. Yet, there are several design problems with the standard mouse-trap. It was designed to be manufactured with simple wire-forming technology. While the manufacture is completely automated, there are nine separate components to be manufactured and assembled (refer to Figure 6.4). Setting of the trap is fraught with some degree of risk to the fingers. If the spring is too strong, the trap can almost decapitate the mouse, and cleaning the trap subsequently can be unpleasant and often results in simply discarding the trap together with its victim.

Figure 6.5 shows a new design for the mouse-trap. Made mainly from moulded polymer, it has only four components. It is easy to set, with a single

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motion. It is also easy to clean, if necessary, by placing it in a dishwasher. While we can not vouch for its capturing effectiveness, it is, at least from the manufacturing and ergonomic points of view, a *better mouse-trap*. Of course, there are many other possible embodiments for catching mice. Our intention in this design demonstration was to compare these two traps with similar *design intent*.

The summary design procedure for the mouse-trap involved:

- identification of a *need*: recognising that there were some manufacturing and ergonomic problems with the original design, leading directly to the *design goal* of creating a better mousetrap;
- setting *technical objectives*, based on experience with manufacturing processes and some wisdom in ergonomics and kinematics: fewer components, easier manufacturing procedure, ergonomic improvements;
- identifying *requirements* within which the embodiment design can survive successfully: material properties and total cost are examples;
- creating *solution alternatives* for the embodiment;

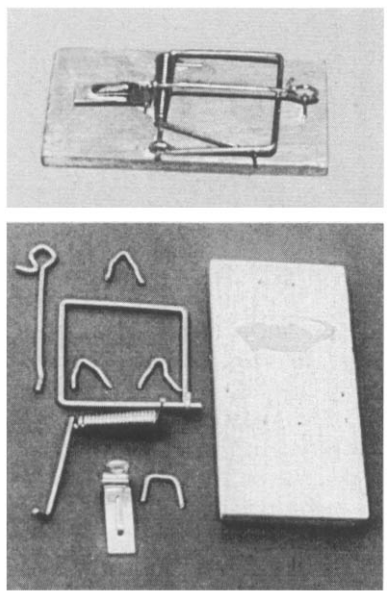


Figure 6.4 Traditional mouse-trap and its component parts

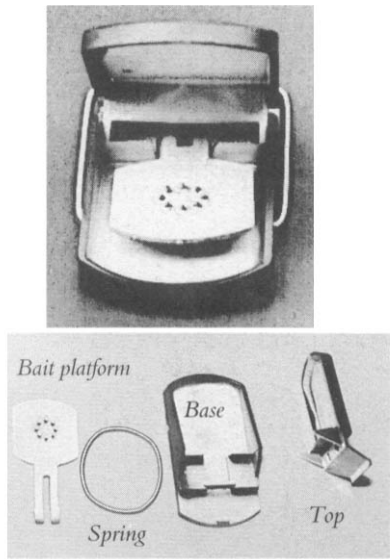


Figure 6.5 New design for mouse trap

- *making decisions* about choice of embodiment: based on evaluating materials, appearance, ease of use, relative cost;
- delivering *design specifications*: including drawings, and material identification and testing.

These two examples of design, one technically complex, the other simple, illustrate the generic design procedure that governs all designs.

Figure 6.6 shows six further examples of consumer product designs, all of which represent a design response to some need. All of them were evolved through the generic process described for the MDX helicopter and the mouse trap. Each is the result of considerable experience, or wisdom, associated with some specific domain of knowledge. Most of the designs in Figure 6.6 are of the type we refer to as *evolutionary*. In these designs, there is still considerable innovative content, but it is usually the result of past experience evolving into some future improvement of an already accepted product.

By contrast, the Bishop variable ratio steering gear, now almost commonplace in many commercial automobiles, represents a different type of innovation– see Figure 6.6(f). It is also the result of a design response to some need. However, the unusual, and completely new, technical content makes this design *inventive* beyond the mere

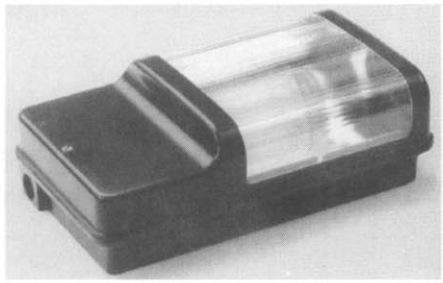
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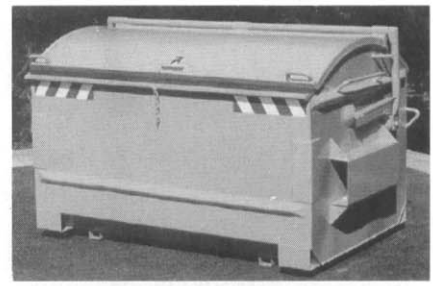
(a) Office chair



(b) Luggage trolleys at an international airport



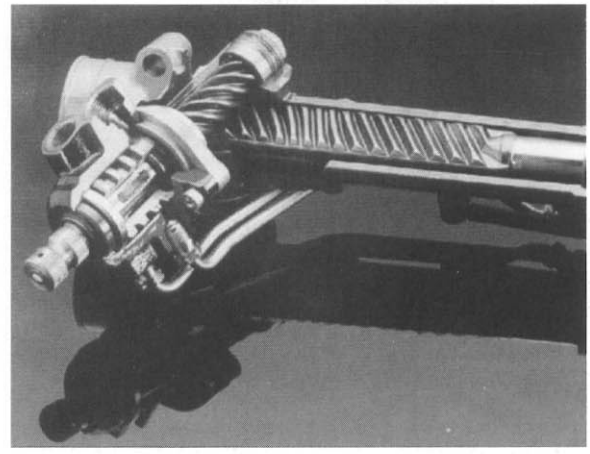
(c) Light fitting



(d) Industrial waste container



(e) Anamorphic printing



(f) Arthur Bishop's variable ratio steering gear

Figure 6.6 Some further examples of consumer product designs

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improvement of an established product. This steering gear renders the steering more direct and responsive at *centre lock*, while driving on a straight path. The steering ratio (turns per degree of steering angle) increases with increasing steering angle, allowing easier parking and manouvering around hard corners.

Figure 6.6 (e)shows a type of printing for flat surfaces that transform images into proper features when the flat surface is deformed into its final container shape. The process relies on a geometric transformation known as *anamorphic* transformation, well known to painters and artists for some centuries (Leeman, 1975). Its application by the container industry is the result of the evolution of new materials and inventive new application of an ancient art. We again see the synergy of experience and wisdom playing key roles in creative design.

6.3 Evolution and enformulation of design problems

6.3.1 The nature of information and problems

In all the examples considered so far, we have dealt with *hard* information. In each case we had some easily identified technical objectives to meet. In all cases the designer may have retained significant creative freedom in the choice of embodiment, but a somewhat restricted freedom to choose a design concept in response to a need. In the case of the *mouse-trap* we didn't consider chemical or electronic or perhaps ultrasonic alternatives to the *real need* : namely a cheap and effective eradication of rodents. We refer to the process of establishing a suitable design concept and *establishing technical objectives* from a specific *need* as *enformulation*.

Enformulation is the result of substantial technical information processing. It brings into play the designers domain-specific know-how (*received wisdom*), as well as a broad awareness of currently available technology, and economic and manufacturing imperatives. Scientists solve problems by operating on abstract models of the problem. Designers can benefit significantly from *enformulating problems initially at an abstract level*. Rodenacker (1970) notes that the most significant creative impact on a design occurs at the highest level of abstraction. A simple and easy-to-understand *need* requiring design enformulation is illustrated by the following example.

At the entrance to some public buildings the presence of birds (mostly pigeons) on the parapet over the entrance can cause a minor hazard to health and clothing (refer to Figure 6.7). This situation represents a *problem*, which is a state wherein we can perceive a *goal* but not the means of attaining it.

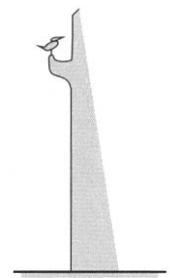


Figure 6.7 Bird hazard at building entrance

Designers (in this case we can think of them as *problem solvers*) respond to problems. Designers (problem solvers) make plans and act on them in ways which they predict will achieve the desired goal.

We identify two categories of problem-solving activity:

(1) planning actions to achieve a goal: we call this *design*;

(2) correcting deviations from desired performance: we call this *trouble-shooting*.

Some engineering texts treat both of these problem solving procedures under the heading of design. Both involve technical problem solving, however, trouble-shooting is more focused on correcting specific, and well-defined, technical problems. Design on the other hand, retains a large element of problem evolution. In what follows, we focus mostly on design problem solving.

As we have noted, the design goals for the products described in the previous section were the result of *hard* information. In contrast, the design goals we might develop for the *bird hazard* would be based on *soft* information. We have no data to tell us how serious the *bird-hazard* is in terms of its impact on eye injuries or dry-cleaning bills. If we had this type of data, we could estimate the comparative value of design effort that could be effectively devoted to solving this problem. After all, there are many, possibly more worthy, design

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problems making calls on our design capacity. Part of the design *enformulation* involves the *hardening of information*.

The following are some examples of problem statements. Some contain *soft* and others contain *hard* information. Unfortunately, designers are occasionally also confronted with what appear to be problem statements, but contain no information at all. One of the many skills that designers need to develop is the capacity to distinguish between these types of problem statements.

"There are too many road deaths" – soft information

"The road deaths per passenger kilometre on Victorian roads is 5% higher than in all other states" – hard information;

"Students need large desks" – soft information;

"The average student desk top measures 1m x 2m" – hard information;

"Our cities are ugly" – content-free, contains no useful information;

"Our public transport system is inefficient" – content-free, contains no useful information;

"Nobody thinks about the hardships a paraplegic endures" – content-free, contains no useful information;

"Clean air is essential for modern society" – *content-free,* contains no useful information.

Motherhood statements was the old-fashioned way of referring to *content-free* problem statements. We prefer to use the more expressive, gender-neutral, version to indicate that these type of problem statements carry no useful information and in general they are to be avoided.

6.3.2 Problem evolution

In an ideal world we could imagine design proceeding, in some nice serial fashion, from *problem* to *solution*. We could also idealise the role of the designer as some *external agent* to the *problem*, who, once advised of the *problem,* could be relied upon to generate the *solution*, almost without interference from the originator of the *problem*. This is the model of problem solving to which we have become accustomed in solving scientific problems. Additionally we have come to expect that some problems, specifically those in mathematics, have unique solutions.

Real design proceeds through a sometimes protracted series of negotiations between designer and client. This is also the case when the designer and client (originator of the problem) are the same person, perhaps *wearing different hats*. Clearly, when the designer and client reside in the same body, these negotiations may take place with less articulation than otherwise. Nevertheless, the negotiations remains an integral part of the design process.

The first negotiation to be managed is the clear identification of a *design goal*. Most commonly, problems in design are stated in a *solution-oriented* way. Examples are: *build a better mouse-trap*; *design a new refrigerator*.

In these example *problem-statements* a specific style of outcome is embedded by the world-experience of the problem's originator. She/he has already experienced mouse-traps and refrigerators, but found some disagreeable features that might be overcome by a *new design solution*. We will allude to this type of mental delimitation of the problem later in the chapter, but it is important to realise that this type of delimitation imposes the most severe restrictions on our problem solving process.

There are many clichés and aphorisms in the design repertoire that arise from experience with this type of delimited problem statement :

" we need to think outside the square" - a variant on de Bono's notion of lateral thinking;

"polishing the bullock-cart will never lead to the automobile or aeroplane";

"what you want is not necessarily what you need";

"designers are always asked to 'fix it, but don't change a thing'".

A useful way of thinking about problems is to regard them as *need-directed*, or *goal-directed*. A *need-directed* problem description represents a vague, general, dissatisfaction with some aspect of the current world. Additionally, in general, a need-directed problem description has goals which are implicit, or *unarticulated*. The design process resulting from a need-directed problem description is, in general, *open-ended*. The designer and problem originator must evolve, through some form of negotiated procedure, clearly articulated design goals. No design work can proceed without these design goals.

In contrast to need-directed problem descriptions, the goal-directed problem statement has a specific goal (or goals) which is (are) more

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clearly articulated. Both types of problem statement are valid, and have different advantages. While we strongly advise avoiding problem descriptions bound up with some imagined solution embodiment, we recognise that these types of problem descriptions are common. Consequently, the process of negotiation from problem description to manageable design goals can be regarded as either a *forward-chaining* or *backward-chaining* problem solving procedure.

Figure 6.8 shows a generic flow-diagram for evolving design goals from need-directed problem descriptions. We can negotiate with our client to develop acceptable and achievable alternative design goals. This is indicated by the *Level 1* boxes in Figure 6.8. Each design goal will evolve its own set of technical objectives, indicated by the *Level 2* boxes. The sets of technical objectives will lead on to some set of embodiments at *Level 3* of our problem solving process. We could move on to *Level 4* (not shown), representing detailed design specifications, eventually followed by manufacturing specifications, and so on.

A simple, familiar, example will illustrate the procedure. Suppose that we are concerned with purchasing an automobile. The two choices under consideration are:

- a two seater convertible sports coupé: for speed and personal image;
- a medium size, four seater sedan: for conservative urban transport.

At this stage of the decision making process, we have already *forward-chained* to the embodiment stage of our problem solving. We began with a problem description of the form:

my transport arrangements are unsatisfactory.

After some negotiation and evaluation of the various transport alternatives, we have identified the *design goal* as:

I need to select an automobile best suited to my needs.

Our *technical objectives* may have included some specification of the maximum cost, or the minimum passenger accommodation. It is unlikely that we would have considered such matters as minimum rate of acceleration, fuel economy or other such technical performance characteristics. In the current climate of automobile marketing these performance characteristics are very similar for automobiles in similar price brackets.

Hence, we come to the choice of *embodiment*. Our decision problem is embedded in our perception of what each alternative choice might offer as *excitement quality* to our purchase. The *sports coupé* will provide excitement and a special personal image, while the *sedan* will provide conservative urban transport to an eventual family. If we now submit our whole problem evolution process to a *design-review* and ask ourselves:

'what is it I really want out of this purchase?: what is the real essence of the problem?',

then we are essentially *backward chaining* in the problem evolution process.

The result of the *design-review* might lead to our recognising that our *strategic needs* are best served by the *conservative sedan*, and we can still get the *image and excitement* aspects by *joining a sports racing club*. Naturally there may be other *design* alternatives. Whatever the outcome, however, the

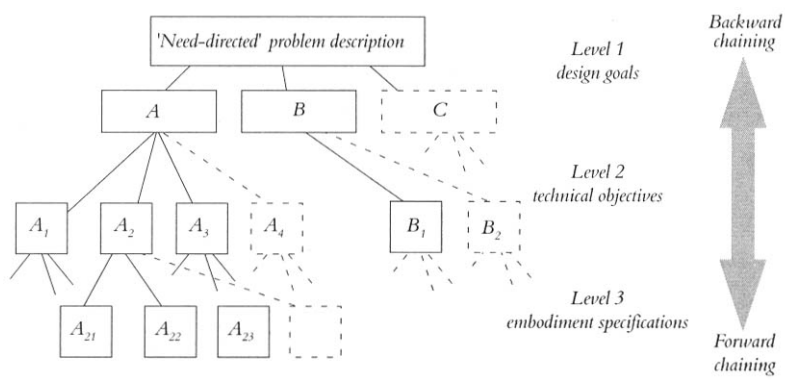


Figure 6.8 Evolving the design problem

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essence of this *backward-chaining* process is that *design reviews* are absolutely essential to the overall health of the problem evolution process.

Before moving on to discuss the enformulation process, we note that problem descriptions can cover a spectrum of styles, ranging from the completely *need-directed* to the completely focused *goal-directed* form. Table 6.1 lists some examples of problem descriptions.

Ex 6.1 The 'bug-list'

An exercise for design teams of two to six people.

In every walk of life we find problem situations that become irksome and eventually grate on our consciousness. We might hear people referring to such situations as:

"this X really bugs me!",

where X represents some irksome problem. Typical examples are:

"I hate to get up on cold mornings";

"there are some really bad smells in my refrigerator";

"there is never enough space on notice boards - nobody clears off old notices";

"I hate looking for parking space at the supermarket".

Historically, the term *it bugs me* is probably derived from the technical vocabulary suggesting *there is a bug in the system*. Radio and other communication systems, relying on early forms of electromagnetic relays, would sufer loss of contact, particularly in the tropics, when small mites, *bugs*, became lodged between the relay contacts. In these cases there really were *bugs* in the system, preventing it from functioning. Nowadays, *"there's a bug in my system"* has become associated with any difficult-to-identify problem that makes a system malfunction. Hence, we have such terms as the *"Y2k bug"*, a reference to malfunctions expected when computer clocks showed two zeroes in the date at the end of the 9th decade of the 20th century.

(a) Make a list of twenty problems that *bug* you. When completed, present your list to your design team, perhaps expanding on each item in your *bug-list*. In essence, you will be expected to *defend* your *bug-list* within your peer-group of designers.

(b) The design team should jointly agree on a suitable *problem-statement*, selected from the several *bug-lists*, to be developed into a sustainable *design goal*. The team should then negotiate their way through to some reasonable *technical objectives*. In each case, the owner/proposer of the chosen *problem* will act as *design client* to the rest of the design team.

(c) Prepare a poster presentation of your chosen *problem*, to be viewed by other design teams. The poster must identify the *problem-statement* and the *bug* from which it was derived, and the resulting *design goal*, in the form of a series of sketches. Your design team should plan the poster presentation to be, as far as possible, self-explanatory.

Notes:

1. In generating the *bug list*, and its distilled *design goal*, your team should keep a record of ideas explored during the design negotiation process, as well as notes and sketches. (We call this an *idea-log*: refer to McKim, 1972. In general, the *idea-log* represents a series of *snap-shots* of the designer's

Table 6.1 Some typical problem descriptions

Problem type	Problem description
Completely need-directed	– emergency vehicle access in the central business district needs to be improved
Order of increasing specific goal focus ↓	– people suffering from arthritis find it difficult to open packaged food: a solution is needed
	– design specifications are needed for an apple harvester
	– design specifications are needed for a domestic washing machine
	– design a horn for an automobile
Specific goal-directed	– design an antenna for a geo-stationary communication satellite

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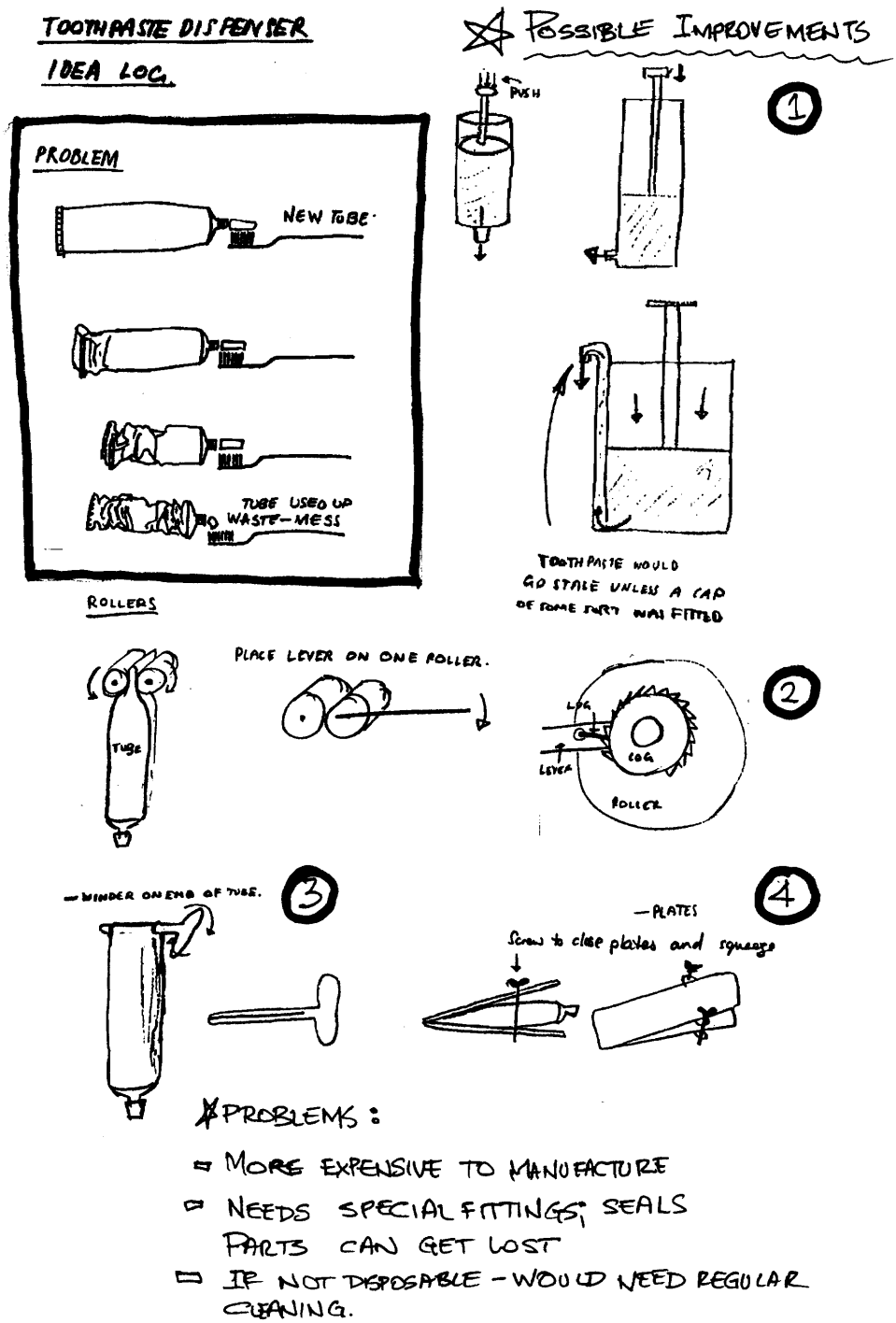


Figure 6.9 Typical idea-log: a problem and some solution alternatives

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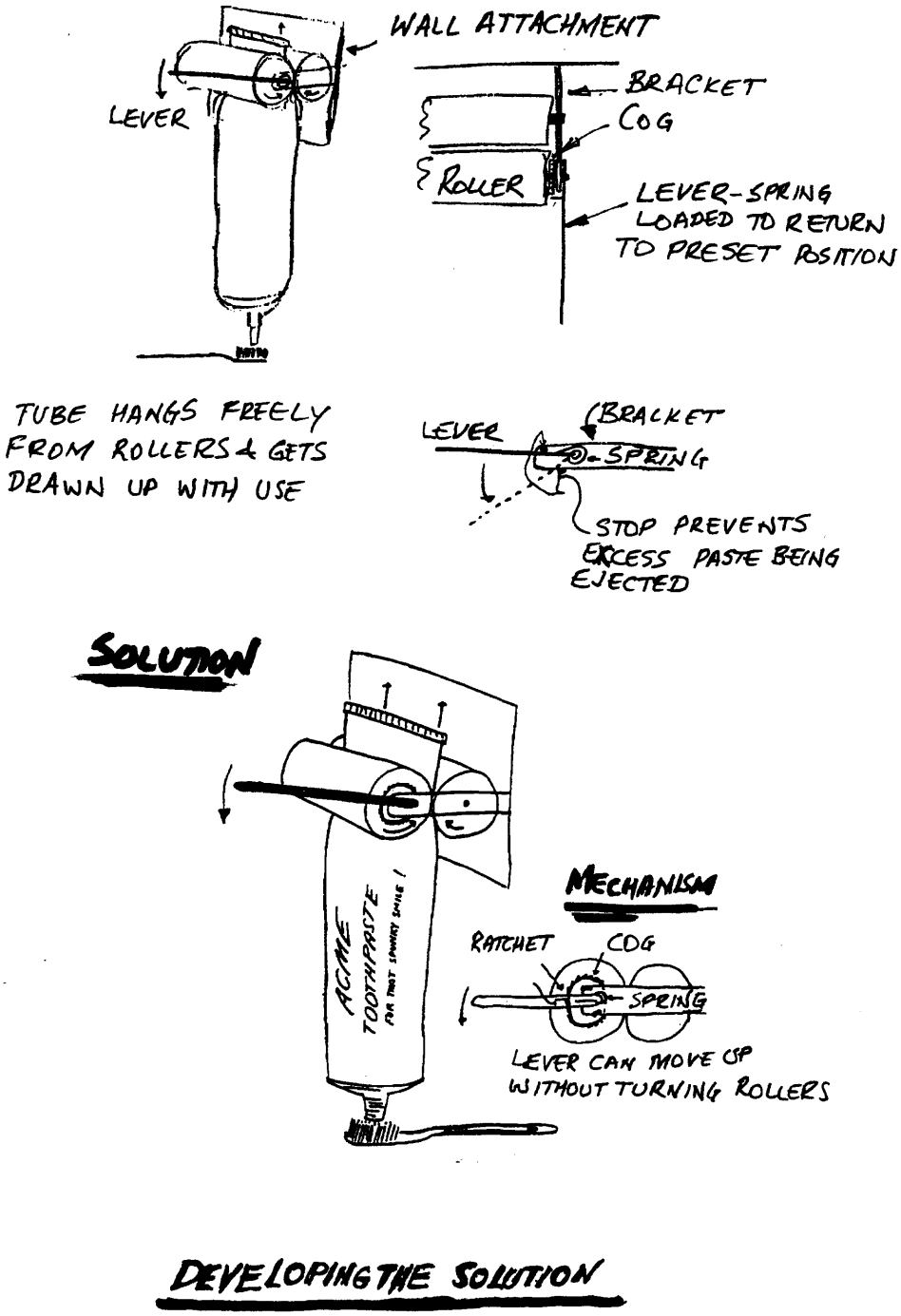


Figure 6.9 (continued) Typical idea-log: developing the solution

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thinking process during problem evolution. Figure 6.9 shows a typical idea-log.)

2. To help you generate your *bug-list*, we offer a list of problem areas you may wish to explore. The list is intended only as a memory-jogger, and you need not adhere to problem areas offered. In each category we note some examples of design goals generated in past bug exercises:

Food *(production, preparation, distribution): keeping hospital food warm; making buttered toast; improved handling of eggs (to reduce breakage);*

Shelter *(mass produced housing or household items):keyless homes; toothpaste dispenser; improved bathroom fixtures; quiet toilet cistern; auto dog food dispenser;*

Clothing *(cleaning devices, disposable items): home dry-cleaner; auto ironing system;*

Transportation *(public or private transport, highways, freeways, rapid transit, congestion): campus/ urban congestion; car escape system; braking-rate indicator; quick windshield de-icer; improved car jack; improved spark plug remover;*

Communications *(written, spoken, telephone, electronic, radio television): improved notice board management; braille communication aid; device to awaken the hearing-impaired; television advertising avoider (to skip over advertising during entertainment broadcasts);*

Recreation *(children's toys, games, sports items, camping/hiking gear): spherical rolling toy (roll in any direction); anti-loss golf ball; improved design for harp tuning;*

Education *(lecture theatres, multi-media, libraries): distributed libraries and learning aids; quiet lecture theatres (means of controlling noise in large classes);*

Environmental *(deforestation, pollution, waterways, urban environment): reduced urban noise; portable pollution meter; trip planner (means of avoiding congested traffic areas);*

Human support *(aids for aged and disabled, access to buildings, hospitals, health and emergency services): anti-damage thermometer case; optical support for micro-surgery; therapeutic exerciser for weakened muscles; special aids for arthritic people (opening packages, turning on faucets etc.);*

3. For reasons that remain a mystery, we find that some student groups engaged in the bug-list exercise become obsessed with scatological concerns – typically: loud toilet flush, cold toilet seats, smelly toilet, seat too low, seat too high and similar. These may be valid design concerns, but try not to limit your thinking in this way.

Ex. 6.2

Figure 6.6 shows several products that had significant design input. In each case, write down a *problem-statement* to which the various design embodiments, shown in Figure 6.6, could be the design responses. Also write down what you consider to be appropriate *design goal*, that might follow from each suggested *problem-statement*. Do your design goals match the design outcomes shown in Figure 6.6? Do you think some other *design goals* might have better served your problem statements?

Note: The review procedure suggested by this exercise is a preliminary *design audit*. We will refer to *design audits* later in this chapter.

6.3.3 Problem enformulation

Part of the designer's enformulation task is to process *soft* information into *harder* information, by identifying *islands of certainty* associated with the problem. Figure 6.10 shows an abstracted model of the design enformulation process.

In the model of the design enformulation shown in Figure 6.10, we see the vaguely stated *problem* mapped onto several sets of technical objectives. As a matter of course, there will be considerable overlap between these sets of technical objectives. Nevertheless, each of these sets of objectives commonly represent distinctly different design alternatives. It is in the nature of design that there is no unique one-to-one mapping between *problem* and *technical objectives*. We have avoided calling these sets of technical objectives *solutions* to the problem since, in general, a solution implies some form of embodiment. Considerable creative design work is involved in moving from sets of technical objectives to the final embodiment. Each of the samples of product design shown in Figure 6.6 represents only one of many possible embodiments that will all satisfy some given set of technical objectives.

We call the process of mapping from *problem* to technical objectives *enformulation*. Enformulation imposes a *structure* on a problem. Additionally, in the enformulation/structuring process we identify

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design parameters that, in general, represent the value measures we use for evaluating the *goodness of fit* of the design solution to our problem. In Figure 6.10 we have used a *curved boundary* to represent the initial *problem-statement*, indicating its relative *softness*. The angular boundaries of the *islands of certainty* within the *soft* boundary of the original *problem-statement* represent established, *hard*, information about the problem. These may be limitations, or *requirements* that bound the problem, or *constraints* imposed by some statutory authority, or even some special *window of opportunity*, that delimits the time-scale or completion time of the enformulation process.

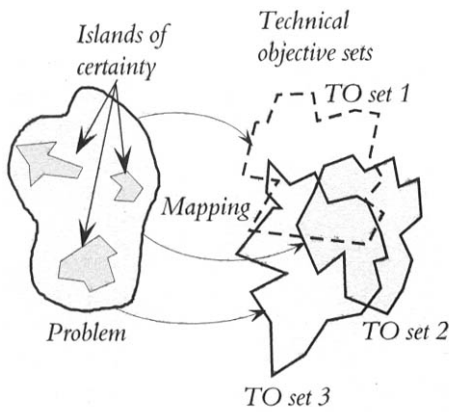


Figure 6.10 Abstraced model of the enformulation process

An example will clarify our terminology.

1. Problem statement (*soft*) : *"Too many bicycles are stolen from the university campus"*
2. *Islands of certainty* : The University of Smallville has 30,000 students, of whom 15% travel to the campus by bicycle. Of these, approximately 4,500 cyclists, two third (3,000) rely on some form of outdoor security for their cycles. The average value of these cycles is $ 500, and approximately 600 to 700 are stolen during the spring term. The total annual loss is approximately $ 325,000. This cost, spread over the cycling population, represents $ 73 annually. If we compare this cost to the $ 400 for car parking annually, or other forms of transport costs, we can see that charging a fee in the vicinity of $ 73 for guaranteed secure parking of cycles is quite reasonable. Moreover, we can consider a three year amortisation of the cost of installation, which is normal business practice for small capital expenditure. Hence we can afford to spend approximately $ 975,000 on installing secure parking for bicycles on the Smallville campus.
3. The layout of the campus is being reviewed, and the buildings administration is preparing a brief for consideration by the finance committee of the university. Plans for minor works on campus will be considered during August this year. This presents a *window of opportunity* for the submission of design recommendations for the provision of secure bicycle parking on campus. A suitable design evaluation must be completed by the end of July.
4. The *requirements* on the design of secure bicycle parking system are:

- it must accommodate at least 1000 cycles – an associated desirable design objective is to deliver a modular design, so that, should the need arise, the ultimate solution may be extended to accomodate up to 4000 cycles;
- it must not cost more than $325,000 per 1000 cycle module to build and install;
- the design plan must match the spread of cycle usage on campus (to be established by survey);
- the design must meet the requirement o f the *Metropolis Standard on Bicycle Parking Facilities*. This a mandatory *constraint* on the design.

CASE EXAMPLE 6.1: LEVEL CROSSING PROBLEM

Level crossings are used where two forms of transport systems, usually road and rail, intersect each other. In sparsely populated Australia, level crossings between road and rail are much more common than viaducts or tunnels. In the early 80's there were several traffic incidents involving property damage, injury, and even deaths at level crossings. Figure 6.11 is an extract from a news article that appeared in the Melbourne Herald in 1981. There was a public outcry for an inquiry to be conducted. What follows is the enformulation

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TRAIN DRIVERS STARE DEATH IN THE FACE

By GEOFF EASDOWN

On the level —it's hell

Train driver Lee O'Reilly . . . a brief delay and the pressure doubles.

HAUNTED BY THOSE FACES

GUIDE TO A CITY'S DEATH TRAPS

'It's there . . . and you hit it'

Jack Vernon

By LAWRENCE MONEY, Herald Transport Writer

Train driver Jack Vernon stood in horror as the white Holden buckled then twisted under his cabin.

Through the window he could see two children being tossed about as the train bulldozed the car down the track.

"I just froze," Mr Vernon said today about the level crossing accident that drove him out of the railways for good.

Melbourne Herald - 1981

Figure 6.11 Extracts from the Melbourne Herald in 1981

of this problem, together with the identification of its *design parameters.*

'NEED-DIRECTED' PROBLEM STATEMENT :

"There are too many deaths and injuries at level crossings."

Note : Major concern was expressed in the press about this problem in 1981. This had the effect of raising public awareness, which in itself may have reduced the problem.

EVOLVING THE PROBLEM : QUANTIFYING ISLANDS OF CERTAINTY:

- how serious is this problem?
- how much is it costing us?
- how much can we spend fixing it?
- where do we put our design effort (and hence expenditure) most effectively?

Table 6.2 shows accident data collected for the years 1982 to 1985.

There are three different types of level crossings, identified by the style of barrier or warning signal used to divide rail traffic from road traffic. Gates and booms are movable barriers, operated either by railway staff, or by automated signals. Flashing lights and bells are used with level crossings without barriers. They provide warning for road traffic users about approaching trains.

The majority of level crossings have no warning signals or barriers, as shown in Table 6.3. However, since these types of crossings are mostly in sparsely populated rural areas, we need to find an appropriate measure for the *seriousness of the problem* in terms of its social or economic impact.

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